SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 5 **Duration : 1 Hour**
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I KG Shalinee (192511048) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
KG shalinee (192511048)

1. AI-Powered Drones for Emergency Medicine Delivery

i. Greedy Best-First Search

Greedy Best-First Search selects the location closest to the destination using the heuristic $h(n)$.

- Strengths: Fast and simple.
- Weaknesses: Ignores actual travel costs and may choose blocked or risky routes.

ii. A* Search

A* considers both the actual cost and estimated remaining cost.

$$ f(n)=g(n)+h(n) $$

Here, $g(n)$ is the cost so far and $h(n)$ is the estimated remaining cost. It can find optimal routes when the heuristic and cost assumptions are satisfied.

iii. Heuristic Accuracy

An accurate heuristic improves route selection. A poor heuristic can increase search time and lead to inefficient routes. A* requires an admissible heuristic to guarantee optimality under standard conditions.

iv. Dynamic Changes

Blocked roads and changing weather require route updates. Both algorithms may need to replan, increasing computation time.

v. Recommendation

A with dynamic replanning* is suitable because it considers travel costs, delays and safety risks. D* Lite may be preferable when routes change frequently.

2. Smart City Traffic Signal Optimization

Formulation

The objective is to minimize waiting time, fuel consumption and congestion.

$$ \min J=w_1T+w_2F+w_3C $$

Traditional search becomes inefficient because the number of possible signal combinations is very large and traffic changes continuously.

i. Hill Climbing

Hill Climbing repeatedly improves the current signal timing plan. However, it may get stuck at a local optimum.

ii. Local Search Problems

- Local optimum: A solution better than nearby alternatives but not the best overall.
- Plateau: Many solutions have similar performance.
- Ridge: Improvement requires coordinated changes across multiple intersections.

iii. Simulated Annealing and Genetic Algorithms

- Simulated Annealing: Sometimes accepts worse solutions to escape local optima.
- Genetic Algorithm: Uses selection, crossover and mutation to explore multiple solutions.

iv. Recommendation

A Genetic Algorithm with real-time traffic feedback is suitable for large-scale optimization. It can explore many solutions and adapt to changing traffic, provided computation is limited appropriately.

3. Autonomous Mars Rover Using Online Search

i. Need for Online Search

Online search is required because the rover explores unknown terrain and cannot prepare a complete route in advance.

ii. Challenges

Partial observations, hidden obstacles, sensor errors and communication delays make navigation difficult.

iii. Internal Model

The rover collects sensor data, updates its map, identifies hazards, estimates movement costs and replans its route.

iv. Search Comparison

- Uninformed search: Uses no heuristic guidance.

- Informed search: Uses heuristics to guide exploration.
- Online search: Learns about the environment while taking actions.

v. Recommendation

Online heuristic-guided search with safety checks is suitable because it adapts to new observations, avoids hazards and supports real-time navigation.

4. University Exam Timetabling Using CSP

Formulation

A Constraint Satisfaction Problem consists of variables, domains and constraints.

i. Variables, Domains and Constraints

- Variables: Exams to be scheduled.
- Domains: Available time slots and examination halls.
- Constraints: No overlapping exams for shared students, hall capacity, faculty availability, subject order and special accommodations.

ii. Backtracking Search

Backtracking assigns a time and hall to each exam. If a constraint is violated, it reverses the assignment and tries another option.

iii. Constraint Propagation

Forward checking removes invalid time slots from other exams' domains after each assignment. This detects conflicts early and reduces search.

iv. Recommendation

Use backtracking with forward checking, MRV and degree heuristics. These techniques improve efficiency and help manage large timetabling problems.

5. Strategic Game AI Using Minimax and Alpha-Beta Pruning

i. Minimax Algorithm

Minimax selects moves by maximizing the AI's score while assuming the opponent minimizes it.

ii. Computational Challenges

With branching factor b and depth d, Minimax has time complexity approximately:

$$O(b^d)$$

Large game trees require significant computation and may exceed real-time limits.

iii. Alpha-Beta Pruning

Alpha-Beta pruning removes branches that cannot affect the final decision. Pruning occurs when:

$$\alpha \geq \beta$$

It reduces unnecessary exploration while preserving the Minimax result for the same searched tree.

iv. Evaluation Function

A simple evaluation function is:

$$E(s)=w_1M+w_2R+w_3T-w_4L$$

Where M is military strength, R is resources, T is territory control and L is unit losses.

v. Recommendation

Use iterative-deepening Minimax with Alpha-Beta pruning. Move ordering and a time limit help the AI make strong decisions quickly.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided